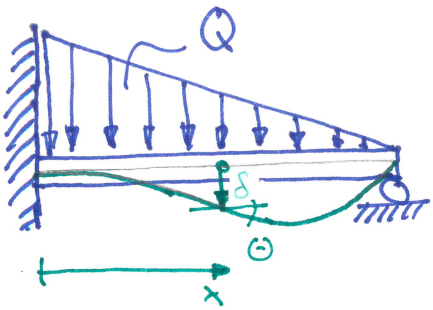
fö 11

Elementarfall

Ex:

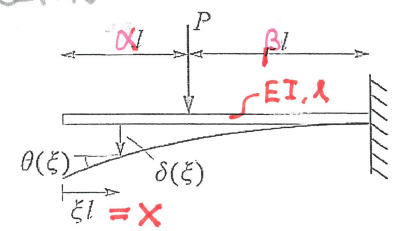
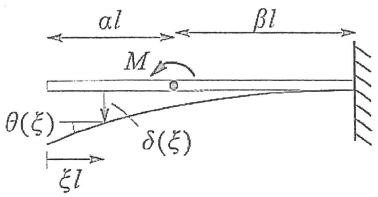


Lösta fall (mha elast. linj. diff.ekv.):
Balkar (L, EI) med olika laster och RV,
map deformationer, stödreaktioner mm.

I handböcker, tabeller, formelsamlingen:

→ 357

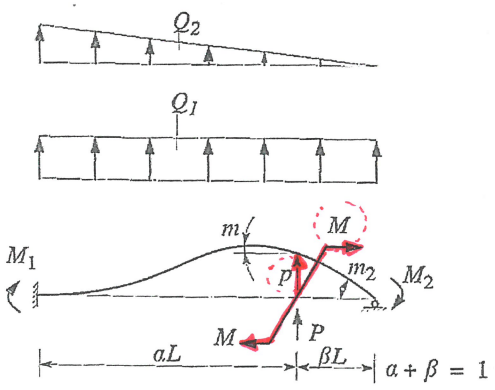
bell 32.1 Elementarfall för konsolbalk

Belastningsfall, $\alpha+\beta=1$	θ = vinkeländring, $\delta(\xi=x/l)$ = förskjutning
Lastens position  Var nånstans?	$\theta(\xi<\alpha)=\frac{Pl^2}{2EI}\beta^2$ $\delta(\xi<\alpha)=\frac{Pl^3}{6EI}(-\beta^3+3\beta^2(1-\xi))$ ← till vänster om P $\theta(\xi>\alpha)=\frac{Pl^2}{2EI}(\beta^2-(\xi-\alpha)^2)$ $\delta(\xi>\alpha)=\frac{Pl^3}{6EI}((\xi-\alpha)^3-3\beta^2(\xi-\alpha)+2\beta^3)$ ← till höger om P $\theta(\alpha)=\frac{Pl^2}{2EI}\beta^2 \quad \delta(\alpha)=\frac{Pl^3}{3EI}\beta^3$
 	$\theta(\xi<\alpha)=\frac{Ml}{EI}\beta$ $\delta(\xi<\alpha)=\frac{Ml^2}{EI}\beta(1-\xi-\beta/2)$ $\theta(\xi>\alpha)=\frac{Ml}{EI}(1-\xi)$ $\delta(\xi>\alpha)=\frac{Ml^2}{2EI}((\xi-\alpha)^2-2\beta(\xi-\alpha)+\beta^2)$ $\theta(\alpha)=\frac{Ml}{EI}\beta \quad \delta(\alpha)=\frac{Ml^2}{2EI}\beta^2$

FS 358-359

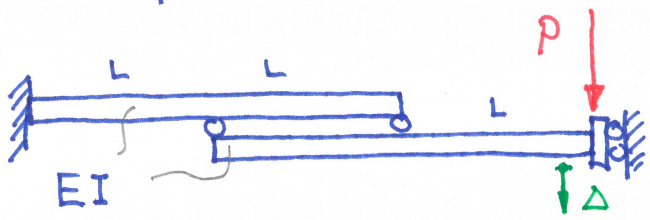
Tex: $P = M \cdot$ [ ] ← proportionalitetskonstant mellan P och M (inkl laster)

Fast inspänd - glidledad balk.



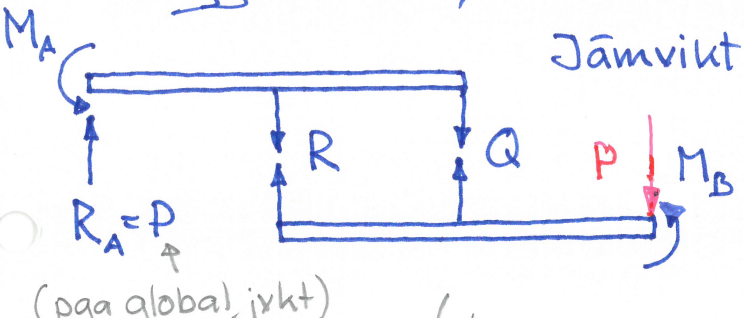
	M_2	M	P	Q_1	Q_2
M_1	$\frac{1}{2}$	$\frac{1}{2}(1-3\beta^2)$	$\frac{L}{2}\alpha\beta(1+\beta)$	$\frac{L}{8}$	$\frac{2L}{15}$
m_2	$\frac{L}{4EI}$	$\frac{L}{4EI}\alpha(1-3\beta)$	$\frac{L^2}{4EI}\alpha^2\beta$	$\frac{L^2}{48EI}$	$\frac{L^2}{60EI}$
m	$\frac{L}{4EI}\alpha(1-3\beta)$	$\frac{L}{4EI}\alpha(\alpha^3+4\beta^3)$	$\frac{L^2}{4EI}\alpha^2\beta(\alpha^2-2\beta^2)$	–	–
p	$\frac{L^2}{4EI}\alpha^2\beta$	$\frac{L^2}{4EI}\alpha^2\beta(\alpha^2-2\beta^2)$	$\frac{L^3}{12EI}\alpha^3\beta^2(3+\beta)$	–	–

Exempel (tentatal): Bestäm styvheten P/Δ.



Notera likformigheten i last och deformation för de två balkarna.

Frilägg delarna, inför snitt- och reaktionslaster:



Jämvikt - Övre: $\uparrow\ P - R - Q = 0$ ①

$\circlearrowleft_A\ M_A - RL - Q\cdot 2L = 0$ ②

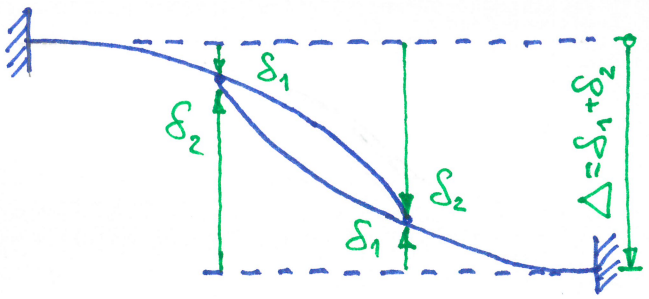
Nedre: $\uparrow\ R + Q - P = 0$ ③ = ①

$\circlearrowleft_B\ M_B - Q\cdot L - R\cdot 2L = 0$ ④

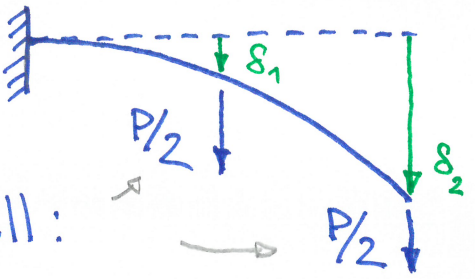
(4 obek, bara 3 ekv. Deformationsvillkor behövs)

Ansätt Δ (Likformighet: Balkdeform. lika, ger $R = Q = \frac{P}{2}$)

och: $M_A = M_B = 3PL/2$

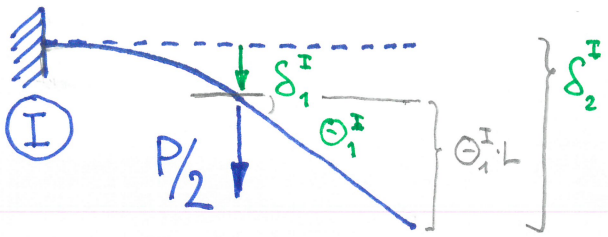


Betrakta (tex) övre



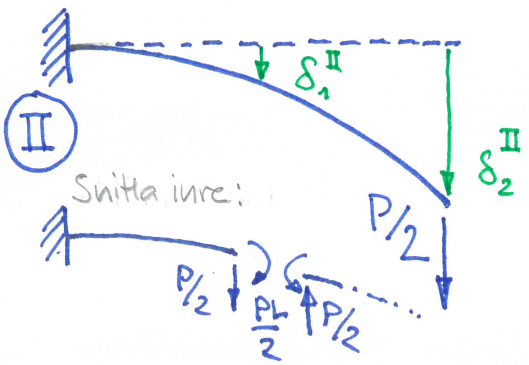
Två lastfall:

Kräv: 4 termer (elementarfall)



$$\delta_1^I = \frac{(P/2)L^3}{3EI} = \frac{PL^3}{6EI}$$

$$\delta_2^I = \delta_1^I + \Theta_1^I L = \frac{PL^3}{6EI} + \frac{(P/2)L^2}{2EI}L = \frac{5PL^3}{12EI}$$



$$\delta_2^{II} = \frac{(P/2)(2L)^3}{3EI} = \frac{4PL^3}{3EI}$$

inre: $\delta_1^{II} = \frac{(P/2)L^3}{3EI} + \frac{(PL/2)L^2}{2EI} = \frac{5PL^3}{12EI}$

pga. kraft moment

Totalt: $\Delta = \delta_1 + \delta_2 = \delta_1^I + \delta_1^{II} + \delta_2^I + \delta_2^{II} = \frac{PL^3}{12EI}(2+5+5+16) = \ldots = \frac{7PL^3}{3EI}$

Varav: $\frac{P}{\Delta} = \frac{3EI}{7L^3}$